

Vachagan Grigoryan

Senior Software Engineer | 6+ years of experience

[vachagan.dev](#) [+374-77-116-777](#) [grigoryan@vachagan.dev](#) [Yerevan, Armenia](#)

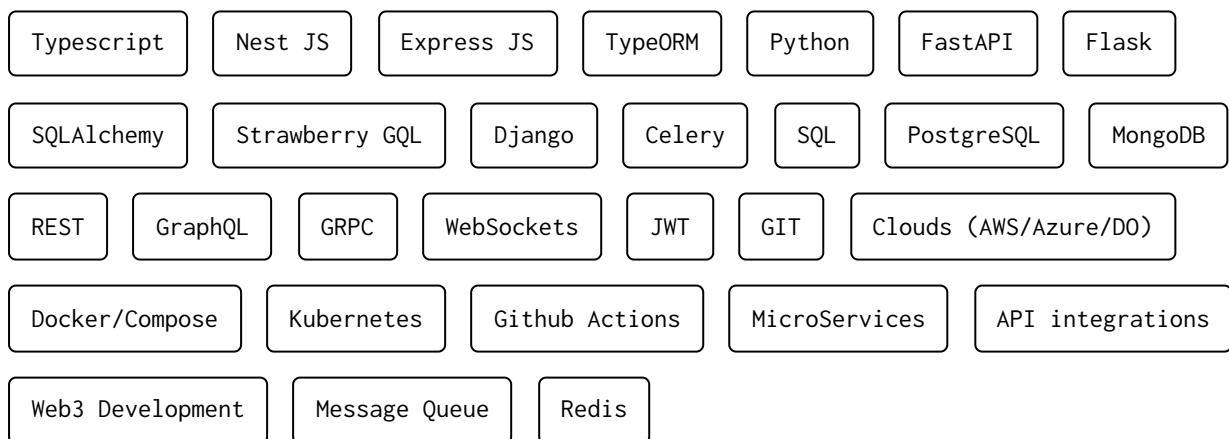
[github.com/VachaganGrigoryan](#) [linkedin.com/in/vachagan-grigoryan](#)

PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience specializing in backend architecture and scalable system design. Expert in Python and TypeScript, with a proven track record of delivering high-impact solutions across the fintech, blockchain, and e-commerce sectors using Django, FastAPI, and Nest.js. Adept in full-stack development with Next.js and Tailwind CSS, and committed to engineering excellence through clean architecture, automation, and the development of performant, maintainable systems.

In addition to software engineering, I've actively contributed to DevOps initiatives—designing and maintaining infrastructure on Azure Cloud, automating deployments with CI/CD pipelines, and managing containerized environments with Docker and Kubernetes. I also built a fully self-hosted system, gaining practical experience in server management, networking, and security best practices.

PRIMARY EXPERTISE



EDUCATION

Bachelor's degree

09/2013 – 06/2017 | Kapan, Armenia

National Polytechnic University of Armenia

Informatics and Applied Mathematics

LANGUAGES

English (Upper-Intermediate)

WORK EXPERIENCE

CTRL Wallet (XDEFI)

03/2022 – 02/2026 | EU/Remote

Senior Backend / Blockchain Developer

Senior Software Engineer leading the design and scaling of distributed microservices and Web3 infrastructure across cross-functional teams.

Achievements:

- Built high-performance microservices using REST, GraphQL, and gRPC to ensure system scalability.
- Spearheaded a high-throughput Blockchain Indexer and core Web3 infrastructure components.
- Architected modular codebases and resolved complex technical bottlenecks to enhance system robustness.
- Optimized CI/CD pipelines and delivery workflows, significantly accelerating deployment efficiency.
- Guided high-level architectural design and product strategy for large-scale backend ecosystems.

Technologies: Python, TypeScript, Go, FastAPI, Flask, Strawberry GQL, NestJS, ExpressJs, Strapi, gRPC, Celery, BullMQ, SQLAlchemy, TypeORM, PostgreSQL, Redis, Firebase, NATS, Jira, Confluence, Linux, AWS, GitHub Actions, Docker, Docker Compose

Zestbloom Inc.

06/2021 – 03/2022 | US/Remote

Python/Django Developer

Led backend engineering for a high-performance Algorand NFT marketplace, architecting scalable API infrastructures and real-time data synchronization services.

Achievements:

- Deployed production-grade RESTful APIs using Django Rest Framework to facilitate seamless NFT minting and high-frequency trading.
- Fortified platform security by implementing custom JWT authentication protocols to safeguard user credentials and API endpoints.
- Optimized asset delivery and reduced latency by integrating DigitalOcean Spaces for high-volume metadata storage and caching.
- Engineered a real-time notification system using Django Channels to provide instantaneous transaction updates and user alerts.
- Developed automated Celery pipelines to parse on-chain data, ensuring precise synchronization between the blockchain and application database.

Technologies: Python, Django, Django REST Framework, Django Channels, Celery, Algorand, DigitalOcean, Docker, Kubernetes, Linux.

Instigate Mobile CJSC

09/2019 – 05/2021 | Armenia

SQA Python Engineer

Led the design and deployment of API automation strategies to enhance software quality and accelerate production delivery cycles.

Achievements:

- Engineered a REST/GraphQL automation framework, significantly increasing test coverage and SDLC efficiency.
- Integrated TestRail, Allure, and Google Sheets to centralize reporting and provide stakeholders with real-time execution insights.
- Developed a JSON mapping utility to automate complex data comparisons, minimizing manual validation errors.
- Optimized test performance and maintainability using OOP and multiprocessing techniques within Linux environments.
- Standardized cross-functional workflows and version control through rigorous Git, Jira, and Postman implementation.

Technologies: Python, Java, PyTest, REST, GraphQL, TestRail, Allure, Google Sheets API, Linux, Git, Postman, Jira, Multiprocessing.

Software Developer

02/2017 – 04/2019 | Remote

Provided full-stack architecture and delivery for a diverse portfolio of cross-industry clients. Engineered scalable web solutions designed to optimize operational efficiency and drive measurable business growth.

Achievements:

- Managed the end-to-end SDLC for high-performance applications, translating complex stakeholder requirements into production-ready systems.
- Developed bespoke full-stack modules that automated critical workflows and significantly increased digital engagement.
- Optimized database schemas and infrastructure to ensure enterprise-grade scalability, security, and data integrity.
- Served as lead technical advisor to align project roadmaps with executive-level strategic business objectives.
- Instituted advanced performance-tuning methodologies and security frameworks to guarantee maximum system reliability.

Technologies: PHP, JavaScript, HTML5, CSS3, WordPress, ModX, jQuery, Bootstrap, MySQL

COURSES

WEB3 Blockchain Course

Yerevan, Armenia

Polytechnic

Python Intermediate Level

Yerevan, Armenia

Armenian Code Academy

Cybersecurity

Yerevan, Armenia

UEICT

Linux, QA, Python

Kapan, Armenia

Instigate Training Center

SELECTED PROJECTS

Web3 Integration

03/2022 – 02/2026

Led the development of scalable microservices for a blockchain-based wallet system, focusing on secure data indexing and decentralized interactions.

Achievements:

- Architected high-performance microservices using REST, GraphQL, and gRPC to facilitate seamless blockchain data handling.
- Engineered secure wallet functionalities by integrating Web3 technologies and maintaining rigorous unit-testing standards.
- Optimized CI/CD pipelines via GitHub Actions, significantly streamlining deployments across distributed environments.
- Managed scalable infrastructure using Docker, NATS, and PostgreSQL to support high-concurrency backend services.
- Contributed to architectural design and product strategy, providing key technical insights for decentralized system integration.

Technologies: Python, TypeScript, Go, FastAPI, Flask, Strawberry GraphQL, NestJS, ExpressJS, gRPC, Celery, BullMQ, PostgreSQL, Redis, Firebase, NATS, Docker, GitHub Actions, AWS

Cloud Integration

11/2024 - 05/2025

Project Description

Worked on integrating and maintaining cloud services for Dutch ICT provider, ensuring seamless operations and scalability. The project focused on cloud infrastructure management, issue resolution, and deployment of new cloud products.

Position

Cloud DevOps Engineer

Technologies

Azure Cloud, Terraform, Kubernetes, Docker, CI/CD, Monitoring Tools, (Prometheus, Grafana), Networking (DNS, SIP), Backup Solutions

Responsibilities

- Maintained and optimized existing integrations on Azure Cloud, ensuring high availability and performance.
- Investigated and resolved cloud integration issues to enhance system reliability.
- Designed and implemented infrastructure for new cloud products, including domains, backup solutions, and SIP services.
- Automated deployments and infrastructure provisioning using CI/CD pipelines.
- Monitored system health and security, implementing necessary optimizations.
- Collaborated with cross-functional teams to align cloud strategies with business goals.

Self-Hosted Home Lab Web Server Setup

2022

Project Description

Designed and deployed a fully self-hosted home lab web server, leveraging a stable internet connection with a static IP and a high-performance custom-built server. The setup provided a secure, automated, and manageable environment for hosting and deploying personal services efficiently.

Position

DevOps Engineer (Personal Project)

Technologies

Ubuntu Server 22.04, Docker, Portainer, GitHub CI/CD, WireGuard VPN, Nginx Proxy Manager, Cloudflare, AdGuard Home, PostgreSQL, Redis, OpenTelemetry/Jaeger

Technical Setup:

- Hardware: Intel i5 (12-core), 16GB RAM, 256GB NVMe SSD, 1TB storage.
- OS & Core Software: Ubuntu Server 22.04 with Docker & Portainer for container management.
- CI/CD & Automation: Integrated GitHub CI/CD for seamless auto-deployment of personal repositories.

Networking & Security:

- Configured private DNS with AdGuard Home for secure and ad-free browsing.
- Set up Nginx Proxy Manager with Cloudflare SSL for secure domain management.
- Deployed WireGuard VPN to restrict access to specific services for enhanced security.

Key Benefits & Achievements:

- Secure & Manageable Self-Hosted Services - Built a private cloud environment with controlled access.
- Quick & Automated Deployments - Hosted essential services like PostgreSQL, Redis, OpenTelemetry/Jaeger for logging and monitoring

- Hands-on Home Lab Experience - Gained in-depth knowledge of self-hosted infrastructure, networking, and deployment best practices.

Blockchain indexer for Binance Beacon chain

03/2022 - 01/2023

Developed a high-performance blockchain indexing and triggering system designed to capture and store real-time block transactions and address balances.

Achievements:

- Designed and delivered scalable microservices using REST, GraphQL, and gRPC to process high-volume blockchain data.
- Optimized system robustness and scalability by resolving complex, large-scale technical bottlenecks.
- Engineered testable and efficient back-end code, implementing comprehensive unit tests to ensure high software integrity.
- Collaborated with DevOps to optimize CI/CD pipelines, streamlining deployment workflows and improving delivery cycles.

Technologies: Python, FastAPI, NestJS, Strawberry GraphQL, SQLAlchemy, gRPC, PostgreSQL, Redis, NATS, Linux, Docker

Next generation Digital Media NFT marketplace

05/2021 - 03/2022

Software Engineer at a next-generation NFT marketplace built on the Algorand blockchain to provide high-efficiency, low-fee digital art trading.

Achievements:

- Designed and deployed core REST APIs using Django Rest Framework to facilitate seamless marketplace transactions.
- Strengthened platform security by implementing and customizing JWT authentication protocols.
- Engineered a real-time notification system utilizing Django Channels to enhance user engagement.
- Developed automated blockchain tracking and parsing engines using Celery to sync NFT data with user wallets.
- Optimized asset delivery by integrating DigitalOcean Spaces for scalable media storage and caching.

Technologies: Python, Django, Django REST Framework, Django Channels, Algorand, PostgreSQL, Docker, DigitalOcean, Celery, JWT